Vadim Vadanici

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Education

Imperial College London — London, UK

MEng Biomedical Engineering (Computational Bioengineering Stream) — *Expected 2028*

- Predicted First Class Honours; top of a 120+ cohort in Year 1 Mechanics & Electronics (79% overall, 83% Mechanics).
- Year 3 modules: Statistical Signal Processing & Inference; Image Processing; Computational Continuum Mechanics.

École Active Bilingue Jeannine Manuel — Paris, France

French Baccalaureate, International Option (English) — *2024*

- Highest distinction (17.61/20) — perfect scores in Mathematics, AP Calculus BC, History, English and AP Language.
- Honourable Mention, French Mathematics Olympiad; Concours Général candidate (Philosophy, Physics-Chemistry).

Research Experience

PITOR — Paris Institute for Transplantation & Organ Regeneration — Paris, France

Research Intern — *August 2026–Present*

- Built a local LLM pipeline automating the thematic analysis of 6,000+ free-text responses from kidney-transplant patients, combining structured coding, embedding-based clustering and source-verified quotations.
- Benchmarked frontier LLMs against expert pathologists on Banff rejection diagnosis from lesion scores in 240 biopsies.
- Preparing a first-author manuscript, continuing part-time alongside my degree.

Dario Farina Lab, Imperial College London — London, UK

Undergraduate Researcher — *2025–2026*

- Applied machine learning to decode single-finger movements from a wrist-worn EMG bracelet, with a PhD student.
- Independently built an open-source PyTorch decoder for 28-channel EMG, tested leak-free on 43 people, 17 gestures.
- Trained a compact CNN that reached 80% on unseen people (LDA: 68%), and 88% after a three-repetition calibration.

Okeiro (AI Medtech Start-up, Chronic Kidney Disease) — Paris, France

Summer Intern — *June–July 2025*

- Developed a US market-entry strategy through competitor benchmarking and nephrology market research.

Engineering Projects

Neural Manifold Learning for Brain-Imaging Biomarkers — London, UK

Project Manager (7-member team), supervised by Prof Simon Schultz — *October 2026–Present*

- Year-long project, the lab's first on human fMRI, applying manifold learning to the UK PREVENT Dementia cohort.
- Refining MIMI, an information-theoretic method that infers low-dimensional structure in brain activity.
- Analysing how default mode network manifolds change with age across participants to find Alzheimer's biomarkers.

Interpretability of Multimodal Models in Oncology — London, UK

Supervised by Dr Guang Yang — *November 2026–Present*

- Applying Google's MedGemma, a vision-language model with a medical image encoder, to colorectal-cancer histology.
- Building knowledge-graph RAG over PubMed for LLM-powered hypothesis generation by clinicians and researchers.
- Extending the lab's text-only PharmkGPT to tissue images, tested against standard-RAG and no-retrieval baselines.

Smart Elastic Resistance Bands for Physiotherapy — London, UK

Project Manager (8-member team) — *October 2025–June 2026*

- Designed sensor wristbands that calibrate to each patient's range of motion, count reps on a resistance band and turn prescribed physiotherapy exercises into a video game to improve adherence at home.
- Developed the ReQuest patient app, clinician dashboard and FastAPI backend for the team's Bluetooth-driven game.
- Led the team to the Year 2 Biomedical Engineering Prize for Best Group Project.

Certifications & Skills

Optiver & Imperial College London — Trading Academy *2026*

Month-long course and competition on options theory and market making; implemented trading strategies in Python.

University of Cambridge — Machine Learning & AI for Medicine Summer School (van der Schaar Lab) *2025*

Programming: Python (NumPy, pandas, SciPy, scikit-learn, PyTorch), SQL, MATLAB, Java, Git, LaTeX

Languages: Fluent in English, French and Romanian; German (B1)

Leadership & Extracurricular Activities

- Selected as one of 500 attendees from 5,000+ applicants for Y Combinator Startup School Paris (2026).
- Organiser, Nature Conference *Redefining Healthcare in the Age of AI* (Sorbonne, Paris, 2026).
- Imperial Neuroscience Society — selected for a mentorship programme led by neurotech founders.
- Former school basketball captain (10+ years); Imperial Basketball Team; EDUCAP/CAPSAA inclusive-sport organiser.
- London AI x Science Hackathon (150 of 1,000+ applicants, 2026): used ML to predict battery manufacturing defects.
- Three exhibitions with Duchamp Prize winner Tarik Kiswanson; subject of his work *Vadim* (kadist.org/work/vadim).
- **Interests:** AI, neuroscience, chess, poker, music (DJing, piano), skiing, football, philosophy.